An Analytic Scaling for Cosmic-Shear Data-Vector Emulation

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1 Idea

A neural emulator learns the map from cosmological parameters $\boldsymbol{\theta}_{\rm c}$ to the cosmic-shear data vector $\mathbf{d}(\boldsymbol{\theta}_{\rm c})$. The dynamic range of $\mathbf{d}$ across the prior is large, which makes the map hard to fit. We remove most of that range *analytically*: divide out a fast approximate model and emulate the (flatter) residual.

Concretely, pick a reference ("mid") cosmology and preprocess each data vector element-wise by

$$\tilde{\mathbf{d}}(\boldsymbol{\theta}_{\rm c}) = \mathbf{d}(\boldsymbol{\theta}_{\rm c}) \odot \mathbf{R}(\boldsymbol{\theta}_{\rm c}), \qquad R \;=\; \frac{\xi^{\rm an}(\boldsymbol{\theta}_{\rm mid})}{\xi^{\rm an}(\boldsymbol{\theta}_{\rm c})}, \tag{1}$$

where $\xi^{\rm an}$ is the analytic model below. The network emulates $\tilde{\mathbf{d}}$ (which clusters near the mid cosmology); the physical vector is recovered as $\mathbf{d} = \tilde{\mathbf{d}} \oslash \mathbf{R}$ before the χ^2, so the data covariance and the reported metric are unchanged. Because R is a *ratio*, any cosmology-common factor (e.g. the bulk of the nonlinear boost) cancels: the model only has to capture the cosmology *dependence*.

2 Approximations

1. Limber projection.
2. Each tomographic source bin is a single source plane at the peak of its $n(z)$ (a δ-function $n(z)$).
3. Pure dark energy expansion, $H(z) = H_0$, giving closed-form distances.
4. Linear matter power spectrum (no halofit).
5. Eisenstein–Hu zero-baryon transfer function.
6. Hankel transform to real space approximated by a δ-function at $\ell\theta = 1$; the Limber integral evaluated at the lensing-kernel peak $u_\star$.

None of these need to be accurate: $\xi^{\rm an}$ is only a reference whose cosmology-dependence we divide out.

3 Projection

For a tomographic source pair (i, j) the Limber shear spectrum is

$$C_\ell^{ij} = \int_0^{\chi_{\rm min}} \frac{q_i(\chi)\, q_j(\chi)}{\chi^2}\, P\Big(k = \tfrac{\ell+1/2}{\chi},\, z(\chi)\Big)\, d\chi, \qquad \chi_{\rm min} = \min(\chi_i, \chi_j), \tag{2}$$

with the single-plane lensing efficiency

$$q_i(\chi) = \frac{3}{2}\,\Omega_m \left(\frac{H_0}{c}\right)^2 \frac{\chi}{a(\chi)}\,\frac{\chi_i - \chi}{\chi_i}, \qquad \chi < \chi_i. \tag{3}$$

Under $H(z) = H_0$ the distance is closed form,

$$\chi(z) = \frac{c}{H_0}\,z \equiv D_H\,z, \qquad a(z) = \frac{1}{1+z}, \qquad \chi_i = D_H\,z_{s,i}. \tag{4}$$

4 Linear power spectrum

The matter spectrum follows from the primordial curvature spectrum and the linear transfer. Write the dimensionless primordial spectrum as

$$\Delta^2_{\mathcal{R}}(k) = A_s\,(k/k_p)^{n_s-1}, \tag{5}$$

with k_p the pivot. In the matter era the sub-horizon density contrast is, through the Poisson equation, proportional to the primordial mode $\mathcal{R}(k)$,

$$\delta_m(k,a) = \frac{2}{5}\,\frac{c^2k^2}{\Omega_m H_0^2}\,T(k)\,D(a)\,\mathcal{R}(k), \tag{6}$$

where $T(k)$ is the Eisenstein–Hu zero-baryon transfer function (Appendix A) and $D(a)$ the growth factor. Squaring and converting to the dimensionful spectrum, $P = (2\pi^2/k^3)\Delta^2$,

$$P_{\rm lin}(k,a) = \frac{2\pi^2}{k^3}\left[\frac{2}{5}\,\frac{c^2k^2}{\Omega_m H_0^2}\right]^2 T^2(k)\,D^2(a)\,\Delta^2_{\mathcal{R}}(k). \tag{7}$$

With growth frozen ($H = H_0$ is de Sitter, so $D = \text{const}$, absorbed into the normalization) and at $z = 0$, collecting the powers of k ($k^{-3}\cdot k^4\cdot k^{n_s-1} = k^{n_s}$) gives

$$P_{\rm lin}(k) = \frac{8\pi^2}{25}\,A_s\left(\frac{c^2}{\Omega_m H_0^2}\right)^2 k_p^{1-n_s}\,k^{n_s}\,T^2(k). \tag{8}$$

5 Amplitude cancellation

The lensing kernel and the power spectrum carry opposite powers of the amplitude parameters. From Eqs. (3) and (8),

$$q_iq_j \propto \Omega_m^2\left(\frac{H_0}{c}\right)^4, \qquad P_{\rm lin} \propto A_s\,c^4\,\Omega_m^{-2}H_0^{-4}\,k^{n_s}T^2(k), \tag{9}$$

so in the Limber integrand the amplitude factors cancel term by term,

$$q_iq_j\,P_{\rm lin} \propto \Omega_m^2\left(\frac{H_0}{c}\right)^4\cdot A_s\,c^4\,\Omega_m^{-2}H_0^{-4}\,k^{n_s}T^2(k) = A_s\,k^{n_s}T^2(k). \tag{10}$$

Hence C_ℓ is proportional to A_s alone: Ω_m and H_0 cancel out of the *amplitude*,

$$\boxed{C_\ell = A_s\,G(\ell;\Omega_m,H_0,n_s,z_s)}\,. \tag{11}$$

In the (A_s,Ω_m,h) parametrization, A_s therefore carries the entire lensing amplitude; Ω_m and H_0 enter only through the *shape* G (the transfer function and the geometry). This is why dividing by $A_s^{\rm mid}/A_s$ is exact and removes the single largest direction of variance.

6 Real space and the effective wavenumber

The correlation functions are Hankel transforms of C_ℓ,

$$\xi_\pm(\theta) = \frac{1}{2\pi}\int_0^\infty \ell\, C_\ell\, J_{0,4}(\ell\theta)\, d\ell. \tag{12}$$

Approximate the oscillating Bessel kernel by a δ-function at its argument, $J_{0,4}(\ell\theta) \approx a\,\delta(\ell\theta - 1)$ for a constant a. Since $\delta(\ell\theta - 1) = \delta(\ell - 1/\theta)/\theta$, the integral collapses onto a single multipole $\ell_\star = 1/\theta$,

$$\xi_\pm(\theta) \approx \frac{a}{2\pi}\int_0^\infty \ell\, C_\ell\, \frac{\delta(\ell - 1/\theta)}{\theta}\, d\ell = \frac{a}{2\pi}\,\ell_\star^2\, C_{\ell_\star}, \qquad \ell_\star = \frac{1}{\theta}. \tag{13}$$

The Limber integral (2) is collapsed the same way, at the lensing-kernel peak $\chi_\star = u_\star \chi_s$, fixing one wavenumber

$$k_\star = \frac{\ell_\star}{\chi_\star} = \frac{1}{\theta\, u_\star \chi_s} = \frac{H_0}{c\, u_\star\, \theta\, z_s}, \tag{14}$$

where the last step used $\chi_s = D_H z_s = (c/H_0) z_s$ from Eq. (4). The transfer function only ever sees the dimensionless combination $q \equiv k_\star \Theta^2/(\Omega_m h^2)$; with $H_0 = 100\,h\ \mathrm{km\,s^{-1}Mpc^{-1}}$ the factor $H_0/(\Omega_m h^2) = 100/(\Omega_m h) = 100/\Gamma$, so

$$\boxed{q \equiv \frac{k_\star\,\Theta^2}{\Omega_m h^2} = \frac{K}{\theta\, z_s\,\Gamma}, \qquad K = \frac{100\,\Theta^2}{c\, u_\star}, \qquad \Gamma \equiv \Omega_m h, \qquad \Theta \equiv \frac{T_{\mathrm{CMB}}}{2.7\,\mathrm{K}},} \tag{15}$$

with θ in radians. The shape parameter $\Gamma = \Omega_m h$ and the source redshift enter *only* through the single combination $q \propto 1/(\theta\,\Gamma\, z_s)$: a rescaling of the angle is equivalent to a change in Γz_s.

7 The scaling

Put the pieces together. With the amplitude stripped (Eq. 11), the Limber spectrum (2) is

$$C_\ell = A_s\, c_0 \int_0^{\chi_s} \frac{d\chi}{\chi^2} \left[\frac{\chi}{a}\, \frac{\chi_s - \chi}{\chi_s}\right]^2 k^{n_s} T^2(k), \qquad k = \ell/\chi, \tag{16}$$

with c_0 a constant. Substitute $u = \chi/\chi_s$, so $\chi = u\chi_s$, $z = u z_s$, $a = 1/(1 + u z_s)$, and $d\chi = D_H z_s\, du$. The geometric bracket simplifies,

$$\frac{1}{\chi^2}\left[\frac{\chi}{a}\, \frac{\chi_s - \chi}{\chi_s}\right]^2 = \frac{\left[u\chi_s(1 + u z_s)(1 - u)\right]^2}{(u\chi_s)^2} = (1 + u z_s)^2 (1 - u)^2, \tag{17}$$

so $C_\ell = A_s\, c_0\, D_H z_s \int_0^1 du\, (1 + u z_s)^2 (1 - u)^2\, k^{n_s} T^2(k)$. Collapse this integral at the kernel peak $u_\star$ and use $\xi_\pm \propto \ell_\star^2 C_{\ell_\star}$ with $\ell_\star = 1/\theta$ (Eq. 13):

$$\xi_\pm(\theta) \propto A_s\, D_H z_s\, (1 + u_\star z_s)^2 (1 - u_\star)^2\, \ell_\star^2\, k_\star^{n_s}\, T^2(k_\star). \tag{18}$$

At fixed θ the factor $\ell_\star^2 = \theta^{-2}$ and the $u_\star$ constants are common to every cosmology and cancel in the ratio R below, so we keep only the cosmology-dependent factors. Splitting $k_\star^{n_s} = q^{n_s} k_\Gamma^{n_s}$ with $k_\Gamma = \Omega_m h^2/\Theta^2$ (so $k_\star = q\, k_\Gamma$) and using $D_H = c/H_0 \propto 1/h$,

$$\xi_\pm(\theta) = A_s\, \mathcal{N}(\Omega_m, h, z_s, n_s)\, q^{n_s}\, T^2(q), \qquad \mathcal{N} \propto \frac{z_s\, (1 + u_\star z_s)^2\, (\Omega_m h^2)^{n_s}}{h}, \tag{19}$$

where $\mathcal{N}$ is a slowly varying geometric amplitude and $q^{n_s} T^2(q)$ is the dimensionless shape.

8 The preprocessing ratio

In the ratio, the survey-fixed parts of $\mathcal{N}$ (z_s and the geometric factor $(1+u_\star z_s)^2$) are identical for the reference and the target and cancel exactly. What survives is a per-cosmology *scalar* amplitude $(\Omega_m h^2)^{n_s}/h$: the Ω_m–H_0 amplitude that cancelled out of the leading C_ℓ (Eq. 11) re-enters here through the tilt $k_\Gamma^{n_s} = (\Omega_m h^2)^{n_s}$ and the $1/h$ distance measure. It is a *second* amplitude direction, independent of A_s, and is kept. Equation (1) becomes the per-element factor

$$\boxed{R(\theta) = R_{\text{amp}} \frac{q_{\text{mid}}^{n_s^{\text{mid}}}\, T^2(q_{\text{mid}})}{q^{n_s}\, T^2(q)}, \qquad q = \frac{K}{\theta\, z_{\text{eff}}\, \Gamma},} \tag{20}$$

with the θ-independent amplitude factor

$$R_{\text{amp}} = \frac{A_s^{\text{mid}}}{A_s} \frac{(\Omega_m^{\text{mid}} h_{\text{mid}}^2)^{n_s^{\text{mid}}}/h_{\text{mid}}}{(\Omega_m h^2)^{n_s}/h}, \tag{21}$$

and $q_{\text{mid}} = K/(\theta\, z_{\text{eff}}\, \Gamma_{\text{mid}})$. Each cosmology uses its own $(A_s, n_s, \Gamma, \Omega_m, h)$ and the reference its own; $R = 1$ at the reference. The θ-dependence enters only through the non-power-law transfer ratio $T^2(q_{\text{mid}})/T^2(q)$ (the genuine shape correction), since $q_{\text{mid}}/q = \Gamma/\Gamma_{\text{mid}}$ is itself θ-independent.

Tomographic cross-pairs. The cross spectrum (2) is cut off at the nearer source, so its effective lensing scale is that of the lower-redshift bin. The single-scale collapse therefore uses

$$z_{\text{eff}} = \min(z_{s,i},\, z_{s,j}), \tag{22}$$

not the geometric mean; for auto-pairs ($i = j$) they coincide. Since z_{eff} cancels at leading order in R, this choice only shifts the (mild) transfer correction, mattering at the few-percent level for asymmetric pairs.

9 Empirical validation

On a held-out validation set, measuring per element the spread ratio across cosmologies $\sigma[\tilde{d}_e]/\sigma[d_e]$ (its square is the fraction of variance *remaining*) and the fraction of elements whose spread the rescaling reduces:

	median ratio	mean ratio	frac. improved
A_s only	0.79	0.76	1.00
+ shape	0.52	0.60	0.89
$+\mathcal{N}$	0.46	0.50	0.98

The exact A_s rescaling alone removes ∼37% of the variance; the shape term roughly doubles that (∼74%); the $\mathcal{N}$ amplitude takes it to ∼79%. Crucially $\mathcal{N}$ also raises the improved fraction from 0.89 to 0.98. The shape term alone leaves the $(\Omega_m h^2)^{n_s}/h$ amplitude in the residual, which dominates at large θ, where $T \to 1$ and the data variation is nearly pure amplitude—exactly the elements the shape term made worse. Restoring $\mathcal{N}$ removes that offset, and it is most reliable there (large scales, no transfer curvature). The reduction is largest at small θ (the well-constrained, small-error-bar scales) and falls to ∼1 only at the largest θ, which carry the biggest covariance and are down-weighted in the χ^2.

10 Validity and limitations

- **Baryons.** The zero-baryon $T(k)$ has *no* Ω_b dependence, so R cannot remove the $\Omega_b h^2$ (physical baryon density) direction. That is precisely the direction that drives the hard, high-$\Omega_b h^2$ cosmologies, so this scaling flattens the broadband bulk but leaves the baryon tail to the network. Capturing it would require the wiggle (with-baryon) Eisenstein–Hu transfer.
- **Nonlinearity.** The model is linear, but R is a ratio: the cosmology-common nonlinear boost largely cancels, so the rescaling remains accurate at small scales (where it empirically performs best) as long as that boost varies weakly across the prior.
- **Large scales.** At large θ the data variation is nearly pure amplitude ($T \to 1$); the $\mathcal{N}$ scalar removes it, so these elements flatten once it is included (the improved fraction rises to 0.98). The small residual sits in the highest-covariance bins and is down-weighted in the χ^2.

A Eisenstein–Hu zero-baryon transfer function

With $\Theta = T_{\mathrm{CMB}}/2.7\,\mathrm{K}$ and k in Mpc^{-1},

$$q = \frac{k\,\Theta^2}{\Omega_m h^2}, \tag{23}$$

$$L = \ln(2e + 1.8\,q)\,, \tag{24}$$

$$C = 14.2 + \frac{731}{1 + 62.5\,q}, \tag{25}$$

$$T(k) = \frac{L}{L + C\,q^2}. \tag{26}$$